

Topic 4.2

4.2: Dynamically Allocating Memory: new and delete

To manage dynamically allocated memory in C++, we may use `malloc` and `free` as in C. However, we have new operators which are somewhat easier to use and also cause some things to happen automatically.

The `new` operator (not a function!) is used to dynamically allocate memory, and returns a pointer to this memory. If the allocation fails, the pointer returned will be `NULL`.

To deallocate dynamic memory, use the `delete` operator.

Examples:

```
int      *p;
int      array_size = 174;
int      *dynamic_array;
Distance *d1, *d2;

p = new int;
if (p)
{
    // Allocation succeeded.
    ...
}

dynamic_array = new int[array_size]; // Allocate
                                     // array of
                                     // array_size
                                     // ints.

d1 = new Distance;                  // No-argument
                                     // Constructor called.
d2 = new Distance(3,4.0);           // 2-argument
                                     // Constructor called.

delete p;
delete dynamic_array;
delete d1;
delete d2;
```

Topic 4.3 (<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-4-dot-3>)